Perceptron Example

We only update the weights when a mistake occurs.

The formula for an update is:

$$w_j = w_j + \alpha y x_j$$

Parameters

Initial Weights (w_0, w_1, w_2): $(0, 0, 0)$

Learning Rate (α): 0.5

Inputs (augmented with 1 for bias): $(1, x_1, x_2)$

Points:

- $(0, 0, -1) \implies (1, 0, 0, -1)$
- $(0, 1, +1) \implies (1, 0, 1, +1)$
- $(1, 0, +1) \implies (1, 1, 0, +1)$

Activation: $f(z) = +1$ if $z > 0$, and -1 if $z \leq 0$

Epoch 1

Step 1: Input (1, 0, 0), Target: -1

Calculate activation function input: $z = (0 \cdot 1) + (0 \cdot 0) + (0 \cdot 0) = 0$

Predict via activation function: $f(0) = -1$

Compare to ground truth: Correct. No update.

Weights: $(0, 0, 0)$

Step 2: Input (1, 0, 1), Target: $+1$

Calculate activation function input: $z = (0 \cdot 1) + (0 \cdot 0) + (0 \cdot 1) = 0$

Predict via activation function: $f(0) = -1$

Compare to ground truth: Error.

Update: $W_{new} = (0, 0, 0) + 0.5(+1)(1, 0, 1) = (0.5, 0, 0.5)$

Weights: $(0.5, 0, 0.5)$

Step 3: Input (1, 1, 0), Target: $+1$

Calculate activation function input: $z = (0.5 \cdot 1) + (0 \cdot 1) + (0.5 \cdot 0) = 0.5$

Predict via activation function: $f(0.5) = 1$

Compare to ground truth: Correct.

Weights: $(0.5, 0, 0.5)$

Epoch 2

Step 1: Input (1, 0, 0), Target: -1

Calculate activation function input: $z = (0.5 \cdot 1) + (0 \cdot 0) + (0.5 \cdot 0) = 0.5$

Predict via activation function: $f(0.5) = 1$

Compare to ground truth: Error.

Update: $W_{new} = (0.5, 0, 0.5) + 0.5(-1)(1, 0, 0) = (0, 0, 0.5)$

Weights: $(0, 0, 0.5)$

Step 2: Input (1, 0, 1), Target: $+1$

Calculate activation function input: $z = (0 \cdot 1) + (0 \cdot 0) + (0.5 \cdot 1) = 0.5$

Predict via activation function: $f(0.5) = 1$

Compare to ground truth: Correct.

Weights: $(0, 0, 0.5)$

Step 3: Input (1, 1, 0), Target: $+1$

Calculate activation function input: $z = (0 \cdot 1) + (0 \cdot 1) + (0.5 \cdot 0) = 0$

Predict via activation function: $f(0) = -1$

Compare to ground truth: Error.

Update: $W_{new} = (0, 0, 0.5) + 0.5(+1)(1, 1, 0) = (0.5, 0.5, 0.5)$

Weights: $(0.5, 0.5, 0.5)$

Epoch 3

Step 1: Input (1, 0, 0), Target: -1

$z = 0.5 + 0 + 0 = 0.5$. Predict via activation function: $+1$. Error.

Update: $(0.5, 0.5, 0.5) + 0.5(-1)(1, 0, 0) = (0, 0.5, 0.5)$

Weights: $(0, 0.5, 0.5)$

Step 2: Input (1, 0, 1), Target: $+1$

$z = 0 + 0 + 0.5 = 0.5$. Predict via activation function: $+1$. Correct.

Weights: $(0, 0.5, 0.5)$

Step 3: Input (1, 1, 0), Target: $+1$

$z = 0 + 0.5 + 0 = 0.5$. Predict via activation function: $+1$. Correct.

Weights: $(0, 0.5, 0.5)$

Epoch 4 (verification)

Step 1: Input (1, 0, 0), Target: -1

$z = 0 + 0 + 0 = 0$.

Predict via activation function: -1. Correct.

Weights: $(0, 0.5, 0.5)$

Step 2: Input (1, 0, 1), Target: $+1$

$z = 0 + 0 + 0.5 = 0.5$. Predict via activation function: $+1$. Correct.

Weights: $(0, 0.5, 0.5)$

Step 3: Input (1, 1, 0), Target: $+1$

$z = 0 + 0.5 + 0 = 0.5$. Predict via activation function: $+1$. Correct.

Final weights: $(0, 0.5, 0.5)$